Solutions and discussion for Assignment 1. Below I go into more details for the answer of these questions. I do not expect you all to have gone this much into detail, but I want to make sure that you understand the reasoning behind the answers. For simplicity, we ignore neutrino mixing and assume that all flavor states have the same mass of 0.1 eV.

Question 1. The kinetic energy of neutrinos in the CνB today is roughly equal to the temperature, $E_{\text{kin}} \simeq \frac{3}{2} \times T_{\text{now}} \simeq \frac{3}{2} \times 1.95$ K $\simeq 2.5 \times 10^{-4}$ eV (I am not asking for the 3/2 factor I used here, you can just take $E_{\text{kin}} \simeq T_{\text{now}} \simeq 1.7 \times 10^{-4}$ eV as stated in the question if you want). Clearly, these are non-relativistic neutrinos today, so $E = \sqrt{p^2 + m^2} \simeq m + p^2/(2m)$ and $p \simeq \sqrt{2mE_{\text{kin}}} \simeq \sqrt{2 \times 0.1 \times 1.7 \times 10^{-4}} \simeq 7$ meV $\simeq 7 \times 10^{-3}$ eV. The typical velocity is then $\beta \simeq p/E \simeq 7 \times 10^{-2}$ of the speed of light. You could also have found the velocity based on $\beta = \sqrt{3T/m} \simeq 0.07c$.

Question 2. The flux of neutrinos of all types and flavors is given by $\Phi = n \times v \simeq 330\,(\nu + \bar{\nu})\text{ cm}^{-3} \times 0.07c \simeq 7 \times 10^{11}\,(\nu + \bar{\nu})\text{ cm}^{-2}\text{s}^{-1}$. This is an order of magnitude larger flux than the entire solar neutrino one, but note that the CνB neutrinos are about 10^6 times less energetic than solar neutrinos. Also, while solar neutrinos are almost exclusively produced as ν_e states, the CνB is made up of all neutrino and antineutrino flavors in approximately equal proportions.

Let us see how these compare for real. In reality, the CνB neutrinos have a spectrum of different energies, with an approximate mean energy that you calculated above. So, it makes more sense to consider the differential flux $\frac{d\Phi}{dE_\nu}$ for the CνB and solar neutrinos, which gives the total flux after integration, $\Phi = \int dE_\nu \frac{d\Phi}{dE_\nu}$. Sometimes, we use flux and differential flux interchangeably, but these are technically different quantities.

Fig. 2 plots the *energy-differential-flux*, $E_\nu \times \frac{d\Phi}{dE_\nu}$ as a function of the neutrino energy together with other neutrino sources to show how the CνB is an extremely abundant source of neutrinos at low energies. Few hundred neutrinos per cubic-cm may not seem like a lot, but they are everywhere and they add up to a significant flux, even when compared with anthropogenic sources such as nuclear reactors and particle accelerators.

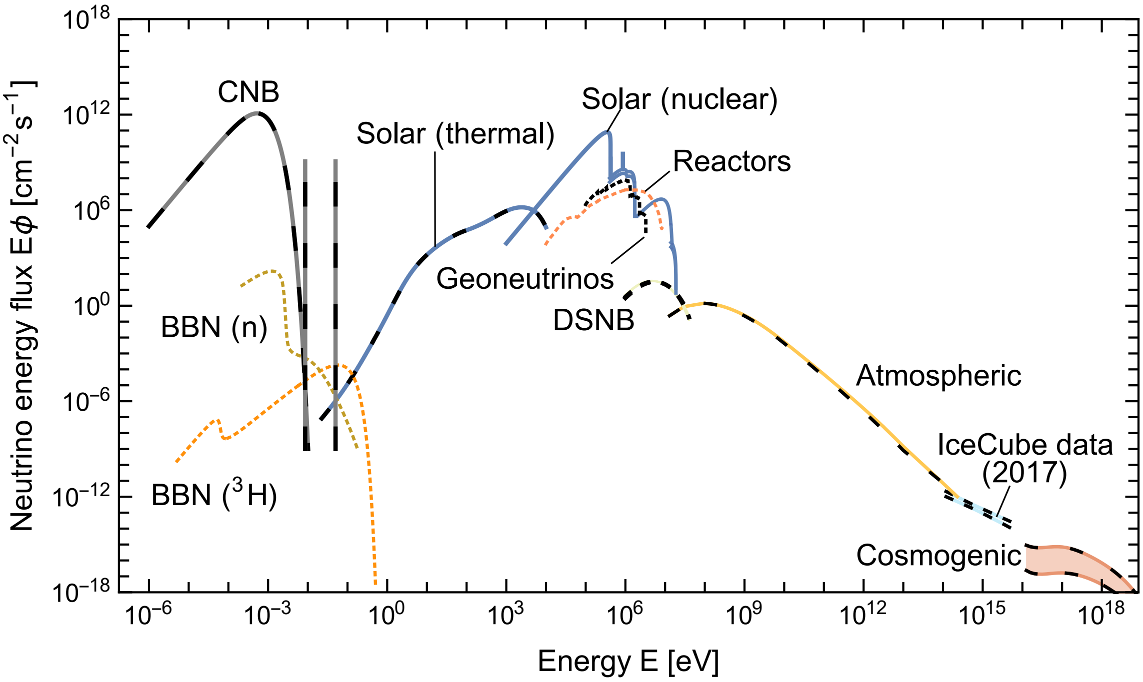


Figure 2: Energy flux of neutrinos from different sources as a function of neutrino energy in logarithmic scale (a more accurate label for the x-axis would be $\log(E_\nu/\text{ eV})$, with tick labels at $-6, -3, 0, 3, \ldots, 18$). The CνB dominates the energy flux at very low energies, while solar neutrinos (nuclear) dominate at higher energies. The spikes appear because this figure accounts for the three different masses of the neutrino mass eigenstates. The two spikes in black-grey dashed indicate that the energy of two of the massive neutrinos are dominated by their mass and that the lightest state is taken to be massless (hence the broad distribution of energies that is allowed to go to zero). Figure from [1].

Why *energy-differential-flux* and not just the differential flux? Well, because we like to plot fluxes in logarithmic scale and maintain the intuition that the area under the curve is the total flux. When making your x-axis logarithmic, the distance between x points is distorted, so we compensate by multiplying the y-axis by E_ν. To see this, consider the total flux that can be calculated as

$$\Phi = \int dE_\nu \frac{d\Phi}{dE_\nu} = \int d(\ln E_\nu)\left(E_\nu \frac{d\Phi}{dE_\nu}\right) = \int d(\ln E_\nu)\left(\frac{d\Phi}{d\ln E_\nu}\right) \\ = \int d(\log_{10} E_\nu)\left(\frac{d\Phi}{d\log_{10} E_\nu}\right), \tag{3.24}$$

by virtue of the change of variables $dE_\nu = E_\nu d(\ln E_\nu)$. So when plotting the flux in logarithmic scale, you can obtain the total flux by integrating the energy-differential-flux, $E_\nu \frac{d\Phi}{dE_\nu} = \frac{d\Phi}{d\ln E_\nu}$ in $\log_{10} E_\nu$ bins.
Take Fig. 2, for example. Because the x-axis is $\log_{10}(E_\nu/\text{eV})$ (replace the label on the figure by what I suggested in the caption), you can find the area under each curve by eye: it is approximately just the value at the peak times the approximate width it spans on the logarithmic x-axis (which is dimensionless! Also, note that the scale on the y-axis is pretty steep, so the peak spans a small width in $\log_{10}(E_\nu/\text{eV})$. In other words, the total flux is about the width in $\ln(E_\nu/\text{ eV})$ (about 0.5 for CνB and maybe $\sim 0.1 - 0.2$ for the solar one).

Question 3. In the Dirac case, we have four degrees of freedom per neutrino flavor: $|\nu_L^\alpha\rangle, |\nu_R^\alpha\rangle, |\bar{\nu}_L^\alpha\rangle$, and $|\bar{\nu}_R^\alpha\rangle$. In the Majorana case, we have only two: $|\nu_L^\alpha\rangle$ and $|\bar{\nu}_R^\alpha\rangle$. In this question, I was just looking for you to write the relations between chiral and helicity states, and then use them to find the proportions of helicity states in the CνB in the past and today. As we will see, some of you made different assumptions about how helicity evolves, so I will discuss these in more detail below. Both are technically correct!

Let us write explicitly the relations between the Dirac neutrino chiral and helicity states for a given flavor using the neutrino velocity $\beta = p_\nu/E_\nu$. We will color code them according to states that interact through the Weak force as a neutrino ($|\nu_L\rangle$) and states that interact through the Weak force as an antineutrino ($|\bar{\nu}_R\rangle$), and sterile states that do not interact at all ($|\nu_R\rangle$ and $|\bar{\nu}_L\rangle$):

Dirac:

$$
\begin{aligned}
|\nu_L\rangle &= \frac{\sqrt{1+\beta}\,|\nu_-\rangle + \sqrt{1-\beta}\,|\nu_+\rangle}{\sqrt{2}}, & |\nu_-\rangle &= \frac{\sqrt{1+\beta}\,|\nu_L\rangle + \sqrt{1-\beta}\,|\nu_R\rangle}{\sqrt{2}},\\
|\nu_R\rangle &= \frac{\sqrt{1-\beta}\,|\nu_-\rangle - \sqrt{1+\beta}\,|\nu_+\rangle}{\sqrt{2}}, & |\nu_+\rangle &= \frac{\sqrt{1-\beta}\,|\nu_L\rangle - \sqrt{1+\beta}\,|\nu_R\rangle}{\sqrt{2}},\\
|\bar{\nu}_L\rangle &= \frac{\sqrt{1+\beta}\,|\bar{\nu}_-\rangle + \sqrt{1-\beta}\,|\bar{\nu}_+\rangle}{\sqrt{2}}, & |\bar{\nu}_-\rangle &= \frac{\sqrt{1+\beta}\,|\bar{\nu}_L\rangle + \sqrt{1-\beta}\,|\bar{\nu}_R\rangle}{\sqrt{2}},\\
|\bar{\nu}_R\rangle &= \frac{\sqrt{1-\beta}\,|\bar{\nu}_-\rangle - \sqrt{1+\beta}\,|\bar{\nu}_+\rangle}{\sqrt{2}}, & |\bar{\nu}_+\rangle &= \frac{\sqrt{1-\beta}\,|\bar{\nu}_L\rangle - \sqrt{1+\beta}\,|\bar{\nu}_R\rangle}{\sqrt{2}}.
\end{aligned}
\tag{3.25}
$$

where we recall that the $_\pm$ subscripts denote the helicity states $h = \pm 1/2$. In this case, it makes sense to separate neutrino and antineutrino states, as they are different particles in the Dirac case.

For Majorana neutrinos, we have only two degrees of freedom per flavor:

Majorana:

$$
\begin{aligned}
|\nu_L\rangle &= \frac{\sqrt{1+\beta}\,|\nu_-\rangle + \sqrt{1-\beta}\,|\nu_+\rangle}{\sqrt{2}}, & |\nu_-\rangle &= \frac{\sqrt{1+\beta}\,|\nu_L\rangle + \sqrt{1-\beta}\,|\bar{\nu}_R\rangle}{\sqrt{2}},\\
|\bar{\nu}_R\rangle &= \frac{\sqrt{1-\beta}\,|\nu_-\rangle - \sqrt{1+\beta}\,|\nu_+\rangle}{\sqrt{2}}, & |\nu_+\rangle &= \frac{\sqrt{1-\beta}\,|\nu_L\rangle - \sqrt{1+\beta}\,|\bar{\nu}_R\rangle}{\sqrt{2}}.
\end{aligned}
\tag{3.26}
$$

Note we do not write any bars on the helicity states since they are a mixture of neutrino and antineutrino interaction states.

In the early Universe, neutrinos were ultra-relativistic, so chirality $\simeq$ helicity. Today, neutrinos in the CνB are non-relativistic, so a chiral neutrino state is an equal mixture of helicity states, and vice-versa. Helicity states are the true propagation eigenstates, so they are the states we care about (see the neutrino-antineutrino conversion example with pion decay in our notes). To understand how the CνB evolved from the early Universe to today, we need to understand how helicity states evolve.

Initially they are produced as ultra-relativistic states with $\beta \simeq 1$, so the overwhelming majority of neutrinos are produced as $|\nu_-\rangle$ and $|\bar{\nu}_+\rangle$ since $\langle\nu_L|\nu_-\rangle = \langle\bar{\nu}_R|\bar{\nu}_+\rangle \simeq 1$ are both order unity, but $\langle \nu_L | \nu_+ \rangle \simeq \langle \bar{\nu}_R | \bar{\nu}_- \rangle \ll 1$ are both negligible (for Majorana, drop the bar from the helicity states). That means that at production, the proportion of states in the CνB is

$$\textbf{Dirac (early):} \quad (|\nu_-\rangle : |\nu_+\rangle : |\bar{\nu}_-\rangle : |\bar{\nu}_+\rangle) \simeq \left(\frac{1}{2} : 0 : 0 : \frac{1}{2}\right), \tag{3.27}$$

$$\textbf{Majorana (early):} \quad (|\nu_-\rangle : |\nu_+\rangle) \simeq (1 : 0). \tag{3.28}$$

The questions asks for the helicity proportions today, in the non-relativistic regime. There are two technically correct answers here depending on the assumptions you make.

Answer 1) Naively, helicity is conserved as neutrinos travel in straight lines since decoupling from the early universe plasma (just like the CMB photons). Their momentum redshifts as $p \propto 1/(1+z)$, but their helicity, $h = \frac{\vec{p}\cdot\vec{s}}{|\vec{p}||\vec{s}|}$, does not change. The only thing that is changing in this case is the chirality content of each helicity state as β decreases. If you made this assumption, the proportions today would be the same as at production:

$$\textbf{Dirac (now):} \quad (|\nu_-\rangle : |\nu_+\rangle : |\bar{\nu}_-\rangle : |\bar{\nu}_+\rangle) \simeq \left(\frac{1}{2} : 0 : 0 : \frac{1}{2}\right), \tag{3.29}$$

$$\textbf{Majorana (now):} \quad (|\nu_-\rangle : |\nu_+\rangle) \simeq (1 : 0). \tag{3.30}$$

Answer 2) However, you could have also assumed that neutrinos do not travel in straight lines. One reason for this would be the gravitational potentials of galaxies that have formed since their decoupling in the early universe. If neutrino velocities are high enough, they keep their helicity as they travel through gravitational potential wells, but if they are slow enough, their helicity can flip since the momentum vector gets deflected but their spin does not. Therefore, an on average, helicity states are fully scrambled today since neutrinos have been traveling through many gravitational potential wells since decoupling and can come from any direction. In this case, the helicity states would be fully mixed today, and the proportions would be

$$\textbf{Dirac (now):} \quad (|\nu_-\rangle : |\nu_+\rangle : |\bar{\nu}_-\rangle : |\bar{\nu}_+\rangle) \simeq \left(\frac{1}{4} : \frac{1}{4} : \frac{1}{4} : \frac{1}{4}\right), \tag{3.31}$$

$$\textbf{Majorana (now):} \quad (|\nu_-\rangle : |\nu_+\rangle) \simeq \left(\frac{1}{2} : \frac{1}{2}\right). \tag{3.32}$$

Okay, but I want to emphasize that in practice this does not make a big difference for the detection of the CνB. To see this, let us take the limit where $\beta \to 0$. In that case, the physical helicity states can interact as $|\nu_L\rangle$ or $|\bar{\nu}_R\rangle$ through the Weak force. For a given helicity state, the probability of interacting as a neutrino or antineutrino is given by the overlap between the helicity and interaction states. Overall, we have

$$\begin{aligned}
&\textbf{Dirac (now):} \\
&P(\nu_-, \nu_L) = |\langle \nu_L | \nu_- \rangle|^2 = \frac{1}{2}, \quad P(\nu_-, \bar{\nu}_R) = |\langle \bar{\nu}_R | \nu_- \rangle|^2 = 0, \\
&P(\nu_+, \nu_L) = |\langle \nu_L | \nu_+ \rangle|^2 = \frac{1}{2}, \quad P(\nu_+, \bar{\nu}_R) = |\langle \bar{\nu}_R | \nu_+ \rangle|^2 = 0, \\
&P(\bar{\nu}_-, \nu_L) = |\langle \nu_L | \bar{\nu}_- \rangle|^2 = 0, \quad P(\bar{\nu}_-, \bar{\nu}_R) = |\langle \bar{\nu}_R | \bar{\nu}_- \rangle|^2 = \frac{1}{2}, \\
&P(\bar{\nu}_+, \nu_L) = |\langle \nu_L | \bar{\nu}_+ \rangle|^2 = 0, \quad P(\bar{\nu}_+, \bar{\nu}_R) = |\langle \bar{\nu}_R | \bar{\nu}_+ \rangle|^2 = \frac{1}{2}, \\
&\textbf{Majorana (now):} \\
&P(\nu_-, \nu_L) = |\langle \nu_L | \nu_- \rangle|^2 = \frac{1}{2}, \quad P(\nu_-, \bar{\nu}_R) = |\langle \bar{\nu}_R | \nu_- \rangle|^2 = \frac{1}{2}, \\
&P(\nu_+, \nu_L) = |\langle \nu_L | \nu_+ \rangle|^2 = \frac{1}{2}, \quad P(\nu_+, \bar{\nu}_R) = |\langle \bar{\nu}_R | \nu_+ \rangle|^2 = \frac{1}{2},
\end{aligned} \tag{3.33}$$

Putting this together with the proportions of helicity states in the CνB today, we can calculate the probability of detecting a CνB neutrino as a neutrino (i.e., through the Weak interaction as a $|\nu_L\rangle$ state):

$$\textbf{Dirac:} \quad P(\mathrm{C}\nu\mathrm{B}, \nu_L) = \sum_{\psi}^{\nu_-,\nu_+,\bar{\nu}_-,\bar{\nu}_+} P(\psi) \times P(\psi, \nu_L), \tag{3.34}$$

$$\textbf{Majorana:} \quad P(\mathrm{C}\nu\mathrm{B}, \nu_L) = \sum_{\psi}^{\nu_-,\nu_+} P(\psi) \times P(\psi, \nu_L), \tag{3.35}$$

where $P(|\psi\rangle)$ is the probability of finding the helicity state ψ in the CνB today, as given in Eq. (3.29) and Eq. (3.31). For Answer 1, this gives

$$\textbf{Dirac:} \quad P(\mathrm{C}\nu\mathrm{B}, \nu_L) \simeq \left(\frac{1}{2} \times \frac{1}{2}\right) + 0 + 0 + 0 = \frac{1}{4}, \tag{3.36}$$

$$\textbf{Majorana:} \quad P(\mathrm{C}\nu\mathrm{B}, \nu_L) \simeq \left(1 \times \frac{1}{2}\right) + 0 = \frac{1}{2}. \tag{3.37}$$

For Answer 2, this gives

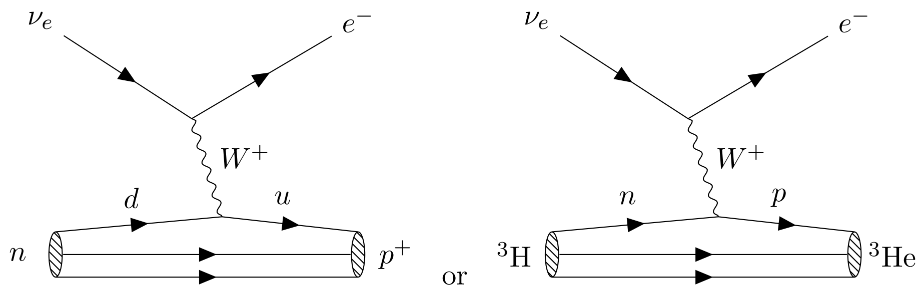
$$\textbf{Dirac:} \quad P(\mathrm{C}\nu\mathrm{B}, \nu_L) \simeq \left(\frac{1}{4} \times \frac{1}{2}\right) + \left(\frac{1}{4} \times \frac{1}{2}\right) + 0 + 0 = \frac{1}{4}, \tag{3.38}$$

$$\textbf{Majorana:} \quad P(\mathrm{C}\nu\mathrm{B}, \nu_L) \simeq \left(\frac{1}{2} \times \frac{1}{2}\right) + \left(\frac{1}{2} \times \frac{1}{2}\right) = \frac{1}{2}. \tag{3.39}$$

A similar calculation holds for the detection as an antineutrino, $P(\mathrm{C}\nu\mathrm{B}, \bar{\nu}_R)$. Therefore, in practice, for the detection of the CνB, it does not matter if all helicity eigenstates are populated (Answer 2) or if only half of them are (Answer 1), as long as neutrinos are non-relativistic today, the combination of all CνB helicity states contains a maximal scrambling of all interaction eigenstates.[7] By the way, the discussion above already answers Question 8, but we will come back to it later.

Question 4. The Feynman diagram for the neutrino capture on tritium is shown below at the quark level (left) and at the nucleon level (right). The initial state neutrino must be a $|\nu_e^L\rangle$ state to conserve charge and lepton number. This means this reaction picks out the left-chiral component of the CνB states, $\langle\nu_e^L|\nu_-\rangle$, etc.

The muon and tau neutrinos cannot be captured by nuclei because there is not enough energy to produce a muon or a tau in the final state. Antineutrinos cannot be captured because lepton number would not be conserved. In principle, you could consider the capture $\bar{\nu}_e^R + {}^3\mathrm{H} \to (3n) + e^+$, but this is not the reaction we were asked about, and, it turns out, it is also energetically forbidden since $3 \times m_n + m_e > m_\nu + m({}^3\mathrm{H})$. You can convince yourself of this by calculating the Q-value of this reaction, as we will do in Question 7 and noting that three neutrons do not form a bound state. While we could benefit from the binding energy of tritium and He3 in the reaction we are interested in, we do not benefit from any binding energy in the hypothetical reaction with three neutrons in the final state. Finally, right-chiral neutrinos cannot be captured because they do not interact through the Weak force.



(Neutrino capture on tritium: $\nu_e + {}^3\mathrm{H} \to {}^3\mathrm{He} + e^-$, mediated by a charged-current W^+ exchange converting a neutron into a proton. The diagram with a W^- is analogous, but the boson mediator is slanted the other way.)

Question 5. The number of neutrino capture events is given by

$$N_{\nu-\text{capture}} = N_T \times \Phi_{\nu_e} \times \sigma_{\nu_e-\text{capture}} \times t, \tag{3.40}$$

where N_T is the number of tritium atoms in the sample, Φ_{ν_e} is the CνB neutrino flux of electron neutrinos, $\sigma_{\nu_e-\text{capture}}$ is the neutrino capture cross section, and t is the time period over which we are counting events. To be more precise, we can build on the question above and say that the flux of electron neutrinos we really care about is

$$\Phi_{\nu_e} = \frac{\Phi}{3} \times P(\mathrm{C}\nu\mathrm{B}, \nu_L^e) \simeq \begin{cases} 0.5 \times 10^{11}\ \mathrm{cm^{-2}s^{-1}}, & \text{Dirac}, \\ 1.0 \times 10^{11}\ \mathrm{cm^{-2}s^{-1}}, & \text{Majorana}, \end{cases} \tag{3.41}$$

[7] Note I am avoiding to say what chirality the CνB neutrinos have today as this is not a well-defined question. Free physical particles have definite helicity, but not chirality. That being said, each helicity state can contain more or less admixture of chiral states.

where the 1/3 factor comes from the fact that the CνB is made up of three neutrino flavors in approximately equal proportions, and $P(\mathrm{C}\nu\mathrm{B}, \nu_L^e)$ is just Eq. (3.33) for electron neutrino states and is 1/4 for Dirac and 1/2 for Majorana neutrinos.

The simplified expression for the cross section can be expressed in more convenient units as

$$\sigma_{\nu_e-\text{capture}} \sim \frac{G_F^2 m_\nu m_p}{2\pi} \simeq 2 \times 10^{-21} \frac{1}{\text{GeV}^2} \simeq 8 \times 10^{-49}\ \text{cm}^2, \tag{3.42}$$

where I used $\hbar c \simeq 2 \times 10^{-5}$ eV.cm $= 1$ to convert from natural units to cgs units. This is a tiny cross section, but the number of tritium atoms in 100 g is quite big:

$$N_T = \frac{100\ \text{g}}{M(^3\text{H})} \times N_A \simeq \frac{100}{3} \times 6.022 \times 10^{23} \simeq 2.0 \times 10^{25}. \tag{3.43}$$

Putting everything together, we still have a small number of events per year:

$$N_{\nu-\text{capture}} \simeq 3 \times 10^{-5}\ \text{events/year}. \tag{3.44}$$

> In reality, this number is much larger(!) because the cross section is significantly enhanced by the fact that tritium is unstable. The capture process is just a stimulated form of beta decay, so it can proceed even for a massless neutrino (hence you could have rightfully be skeptical of the proportionality of m_ν in the cross section). A better estimate for the cross section for neutrino capture on tritium in the limit of massive neutrinos at rest is $\sigma_{\nu_e-\text{capture}} \simeq 1 \times 10^{-42}\ \text{cm}^2 \times \left(\frac{m_\nu}{0.1\ \text{eV}}\right)$ [2], which gives $\mathcal{O}(10)$ events per year for 100 g of tritium when also eliminating other approximations we had to make. This is still quite a small number of events, but at least it is not hopefullylessly small!

Question 6. The lifetime of tritium is about 12.3 years, so in a time period of 1 year, only about $1/12.3 \simeq 8\%$ of the tritium atoms will decay. This is about $\mathcal{O}(10^{24})$ decays, which is a huge number compared to the expected number of neutrino capture events of $\mathcal{O}(10)$ per year.

Question 7. The maximum kinetic energy of the electrons produced beta decay of tritium nuclei is just

$$Q_\beta^{\text{decay}} = \max(E_e - m_e) = \Delta M - m_e - m_\nu = 18.6\ \text{keV} - m_\nu, \tag{3.45}$$

where $\Delta M = M(^3\text{H}) - M(^3\text{He}) \simeq 530$ keV is the mass difference between the tritium and helium-3 nuclei and $m_e \simeq 511$ keV is the mass of the electron (note how $\Delta M > m_e$, as otherwise tritium could only decay via electron capture). In the capture process, the neutrino appears in the initial state, so its mass is injected into the system, increasing the maximum kinetic energy of the electron by m_ν compared to the beta decay process where the neutrino is in the final state.

$$Q_\beta^{\text{capture}} = \Delta M - m_e + m_\nu = 18.6\ \text{keV} + m_\nu. \tag{3.46}$$

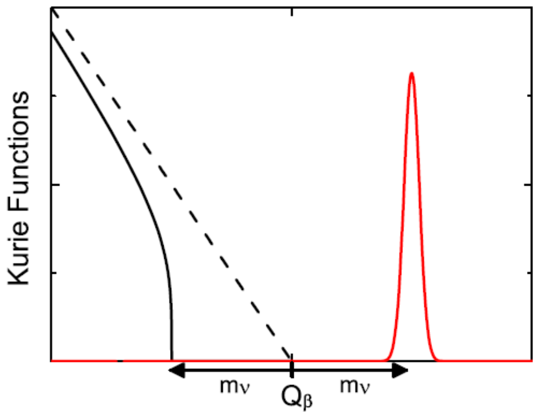


Figure 3: Schematic of the electron kinetic energy spectrum from tritium beta decay (black) and neutrino capture on tritium (red). The two processes are separated by $2m_\nu$ in energy. The finite energy resolution of the detector smears the mono-energetic capture peak. Figure from [3].

So these two processes are separated by (at least) $2m_\nu$ in energy. Tritium decay is a three-body process, so the electron can take a range of kinetic energies up to $Q_\beta^{\rm decay}$, but not beyond $Q_\beta^{\rm capture}$. In capture, the electron is mono-energetic at $Q_\beta^{\rm capture}$ (up to corrections from the motion of the tritium nucleus and of the neutrino). An illustration of the discrimination between the two processes is shown in Fig. 3.

> Why is a 2-body process mono-energetic? To see this, go to the center-of-mass frame, where the two final state particles must have equal and opposite momenta to conserve momentum, so that $|\vec{p}_1| = |\vec{p}_2| \equiv |\vec{p}|$. Their energies are fixed by energy conservation as
>
> $$E_{\rm CM} = E_1 + E_2 = \sqrt{|\vec{p}_1|^2 + m_1^2} + \sqrt{|\vec{p}_2|^2 + m_2^2} = \sqrt{|\vec{p}|^2 + m_1^2} + \sqrt{|\vec{p}|^2 + m_2^2}, \quad (3.47)$$
>
> from which you can show that
>
> $$E_1 = \frac{E_{\rm CM}^2 + m_1^2 - m_2^2}{2E_{\rm CM}}, \quad E_2 = \frac{E_{\rm CM}^2 + m_2^2 - m_1^2}{2E_{\rm CM}}. \quad (3.48)$$
>
> In the lab frame, the energies can vary a bit due to the motion of the initial state particles, but this is a small effect for the case of two non-relativistic targets. In neutrino capture, neglecting the motion of neutrino and the tritium nucleus, $E_{\rm CM} = m_\nu + M(^3{\rm H})$, $m_1 = m_e$, and $m_2 = M(^3{\rm He})$.

Question 8. As we demonstrated in the answer to Question 3, the probability of detecting a CνB (electron) neutrino as a neutrino (i.e., through the Weak interaction as a $|\nu_L\rangle$ state) is 1/4 for Dirac and 1/2 for Majorana neutrinos, regardless of the assumptions we made about how helicity states evolve from the early universe to today. So yes, the rate is different. The underlying reason lies in the number of degrees of freedom and that in the Dirac case, half of the CνB states are sterile and do not interact at all, while in the Majorana case, all helicity states can participate in Weak interactions.

References

[1] E. Vitagliano, I. Tamborra and G. Raffelt, *Grand Unified Neutrino Spectrum at Earth: Sources and Spectral Components*, *Rev. Mod. Phys.* **92** (2020) 45006 [1910.11878].

[2] R. Lazauskas, P. Vogel and C. Volpe, *Charged current cross section for massive cosmological neutrinos impinging on radioactive nuclei*, *J. Phys. G* **35** (2008) 025001 [0710.5312].

[3] Y.-F. Li, *Direct detection of relic active and sterile neutrinos*, *J. Phys. Conf. Ser.* **718** (2016) 062038 [1606.04734].